PPC ENERGIE · AI IN ENERGY, FROM DATA TO ACTION

Free electricity hours, announced before the auction

Decided by our model, acted on by the agent, checked by a human only when it’s unsure.

Mihnea Cucu · Izabela Jilavu · Roland Teslaru · Andrei Oprescu · Ștefan Militaru · 8 November 2026

THE PROBLEM

Which hours can PPC promise customers as free, before the auction clears?

≥ 12 h notice

Customers need to plan washing, charging, heating. The D+1 auction closes at 13:00, so the promise must rest on a forecast, not on prices.

Cancel = cost

A slot that clears expensive must be cancelled. Too many cancellations and customers stop listening.

WHAT GRIDSENSE DOES

It picks the hours PPC can give away as free electricity, tells customers in time, and corrects itself after the auction

Every day, 10:00

Forecast

For each hour of the next 7 days: the chance the price will be near zero (≤ 20 €/MWh).

Right after

Announce

For tomorrow and the day after: sure windows go out to customers on their own; less sure ones wait for an operator's yes.

14:00, auction results

Confirm or cancel

Cleared at or below 0 €/MWh: confirmed. Above 20 €/MWh: cancelled, after the operator approves.

Always

Explain

Every decision shows why, its impact, and answers the operator's questions.

Outputs: customer messages, a calendar of free hours (.ics), alerts for the trading desk and data owners.

THE AGENT'S LOOP

From data to an approved action, every day

Perceive

Pipelines append each day's forecast run and the cleared day-ahead prices to the datasets.

Estimate

Our LightGBM model: calibrated chance each hour clears ≤ 20 €/MWh, plus a P10–P90 price range, D+1 to D+7.

Decide

Rules on the model's numbers, with thresholds from validation. Unsure goes to a human.

Act

Announce, confirm or cancel slots; customer messages, calendar file, team alerts.

DATA

Public data only, every feature known at 10:00 on the issue day

Prices: Romanian day-ahead, hourly (15-min periods averaged).

Weather: forecasts as they were issued days before (archived runs), not observed weather: no look-ahead.

Grid & calendar: load forecasts, holidays, weekends, recent negative-price hours.

Train

Jan 2023 – Feb 2025

Validation: thresholds and calibration chosen here

Mar – Sep 2025 · 35,945 forecast rows

Test: untouched until the single final evaluation

Mar – Oct 2026

Sources: ENTSO-E Transparency Platform · Open-Meteo Previous Runs API · Romanian public holidays.

OUR MODEL VS A SIMPLE BASELINE

LightGBM beats climatology on every metric we decide with

Validation, hour clears ≤ 20 €/MWh	Climatology	LightGBM	Oracle
PR-AUC (main metric, imbalanced)	0.705	0.746	n/a
PR-AUC at D+1	0.705	0.764	n/a
Brier score (lower is better)	0.0855	0.0779	n/a
Precision of the top 70 h / month	0.753	0.792	0.945

Climatology = month × weekday × hour frequencies from training. The oracle knows the real prices: the gap to it is our honest headroom.

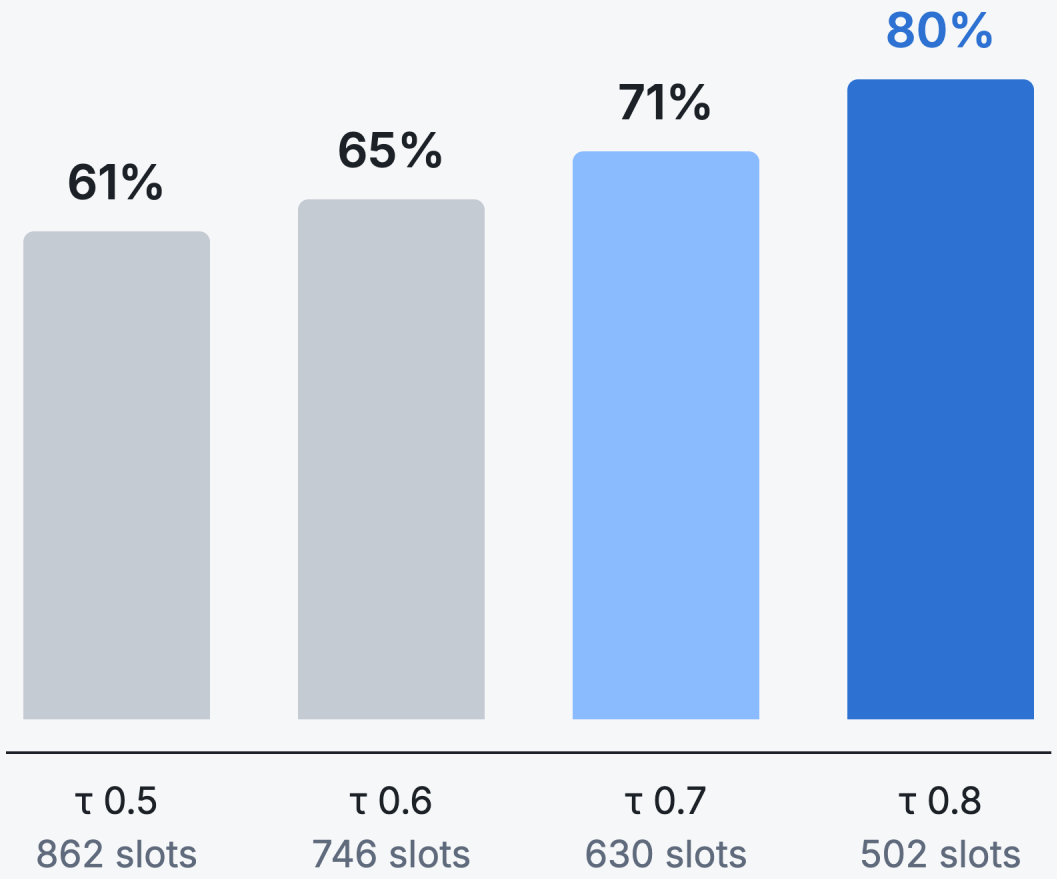
Validation Mar–Sep 2025, 35,945 forecast rows. docs/ml_handoff.md · reports/real/eval/decision_metrics_val.csv

WHY THESE THRESHOLDS

A higher bar means fewer slots, but more of them hold

A slot = 2–6 consecutive hours between 09:00 and 18:00, every hour above the threshold. It “holds” if every hour clears ≤ 20 €/MWh.

τ = 0.8 → Likely, announced automatically.
0.7 ≤ p < 0.8 → Possible, a human decides.



Share of slots that held, calibrated model, validation.
Climatology at τ 0.8: no slot at all.

reports/real/eval/classify_slots_val.csv

HOW IT RUNS

Five services, one flow, no manual step

Pipelines

Python, on cron schedules set in the UI; live logs.

→

Datasets

Forecasts and prices as parquet files.

→

Watchdog

Checks the operator's triggers on new rows; raises hits.

→

Agent

LangGraph: one thread per hit, checkpointed.

→

Operator

Inbox, approve or reject, ask the agent why.

NestJS + BullMQ on Redis, Postgres, React. The same design scales to PPC: features in batch on Spark, scoring daily, events on Kafka.

WHAT AN AGENT IS MADE OF

The trigger says when, the playbook says what, the graph says how

WHEN

Trigger

Conditions on the columns of a dataset, written by the operator in the UI. Every match is a hit.

p_near_zero ≥ 0.7, 09:00–18:00, inputs ok

WHAT

Playbook

What the agent does with the trigger's hits, chosen per trigger.

Announce · Confirm · Cancel · Alert

HOW

Graph

The same steps for every hit, each one logged, paused for a human when unsure.

load → decide → analyze → approval → act

Example: trigger **"Zero-cost window"** + playbook **Announce free hours** = the agent that offers free hours to customers.

THE PLAYBOOKS

Four playbooks today: rules decide, a human approves

Playbook	What the agent does	Used by trigger	Who acts
Announce free hours	Offers D+1/D+2 hours to customers. Likely: 3 h or more in a row at p ≥ 0.8. Possible: every hour p ≥ 0.7.	Zero-cost window	Likely: the agent. Possible: the operator.
Confirm free hours	The auction cleared the hours at ≤ 0 €/MWh: confirms the announced slot, or offers a surprise one.	Negative prices cleared	The agent
Cancel free hours	The hours no longer hold: cancels what was announced and tells customers.	Forecast miss, Forecast dropped	The operator approves
Alert	Tells the trading desk or the data owner; no customer is affected.	Degraded data (template)	The agent

LIVE DEMO · 2 MINUTES

Replaying real days from 24 April 2025, one day per step

1. Advance one day: our model issues the 10:00 forecast, then that day’s auction prices arrive
2. Watchdog: the “Zero-cost window” trigger fires on the new rows
3. Agent: announces a Likely slot on its own, sends a Possible one to the inbox
4. Operator: reads the impact analysis, asks “why?”, approves
5. The auction confirms the slots that cleared at or below 0 €/MWh

Validation period, real prices, the team’s trained models. The test period stays untouched.

ROBUSTNESS, EXPLAINABILITY, RESPONSIBLE AI

When it isn’t sure, it says so and asks

Robust

Missing weather → data marked degraded, never announced on its own. A fresher forecast supersedes older requests. Only D+1 and D+2 decided.

Explained

Every thread keeps its steps, the rule, the numbers it read. Claude writes the analysis and answers questions from read-only tools. It never decides.

EU AI Act ready

Human oversight on customer-facing decisions, a decision log, no personal data sent to the LLM, template fallback offline, a drift metric for the classifier.

IMPACT AND HONEST LIMITS

Fewer broken promises, and we know where the model is weak

Per announced slot (pilot assumptions)

5,000 customers

up to 11.8 MWh shifted

Value = energy bought at the slot's price instead of the evening P50, computed by the agent for each decision.

Limits we found

Cheap hours are over-forecast by 14–32 €/MWh: we trust the classifier, not the price, to pick slots.

Filling the whole 70 h / month budget means ~34% cancellations, so the agent leaves budget unused.

Gap to the oracle: 0.79 → 0.95 precision.

Pilot figures are stated assumptions (central-agent config). Error analysis: docs/ml_handoff.md, validation 2025.

WHAT'S NEXT

Triggers work on any column of any dataset, so a new agent is a new dataset, a trigger and a playbook

New dataset + Trigger on its columns + Playbook = A new agent

Meter anomalies

Rețele Electrice: smart-meter scores → plan tomorrow's inspections.

Asset health

PPC Renewables: turbine and inverter alarms → maintenance tickets.

Trading desk

Evening price spikes and shifted volume → hedging alerts.

Live and 15-min

Live ENTSO-E and weather feeds instead of replay; 15-minute market periods.

Pick the hours. Promise only what holds. Ask a human only when unsure.

GridSense: our model forecasts, rules decide, the agent acts on its own when it’s sure and asks an operator when it isn’t. Every step logged.

Questions